$$P_{\min} = k\frac{S(T_1 - T_2)^2}{lT_1} \quad . \qquad (6)$$

For operating time t and price per kWh c, the least electricity cost is

$$C_{\min} = P_{\min} tc \quad . \qquad (7)$$

Note: $T_1 = 20.00\ \mathrm{K} + 273.15\ \mathrm{K} = 293.15\ \mathrm{K}$, $T_2 = -5.00\ \mathrm{K} + 273.15\ \mathrm{K} = 268.15\ \mathrm{K}$, 1 unit=1kW . Equations (6), (7), with the stated data, give

$$C_{\min} = \frac{(T_1 - T_2)^2}{T_1 l} Sktc = 23.99 \text{ yuan.} \qquad (8)$$

Thus, ideally, this heat pump costs about 24 yuan for 12 .

2. Let T_i, T_0 be the inner and outer air-layer surface temperatures. The rates of heat conduction through the inner glass, air and outer glass are respectively

$$H_1 = k\frac{T_1 - T_i}{l}S \quad , \qquad (9)$$

$$H_2 = k_0\frac{T_i - T_0}{l_0}S \quad , \qquad (10)$$

$$H_3 = k\frac{T_0 - T_2}{l}S \quad . \qquad (11)$$

In steady heat conduction,

$$H_1 = H_2 = H_3 \quad . \qquad (12)$$

Equations (9) ～ (12) give

$$k\frac{T_1 - T_i}{l} = k_0\frac{T_i - T_0}{l_0} \quad \text{and} \quad T_1 - T_i = T_0 - T_2 \quad . \qquad (13)$$

Solving (13) gives

$$T_i = \frac{l_0 k + lk_0}{l_0 k + 2lk_0}T_1 + \frac{lk_0}{l_0 k + 2lk_0}T_2 \quad . \qquad (14)$$

Substitute (14) into 9):

$$H_1 = \frac{kk_0}{l_0 k + 2lk_0}(T_1 - T_2)S \quad . \qquad (15)$$

Constant temperature requires $q = H_1$. By (3), for double glazing the minimum electric power is

$$P'_{\min} = \frac{kk_0}{l_0 k + 2lk_0}\frac{(T_1 - T_2)^2}{T_1}S \quad . \qquad (16)$$

Ideally, the cost for operation time t is

$$C'_{\min} = P'_{\min} tc \quad ; \qquad (17)$$

Substituting the data gives

$$C'_{\min} = 2.52 \text{ yuan.} \qquad (18)$$